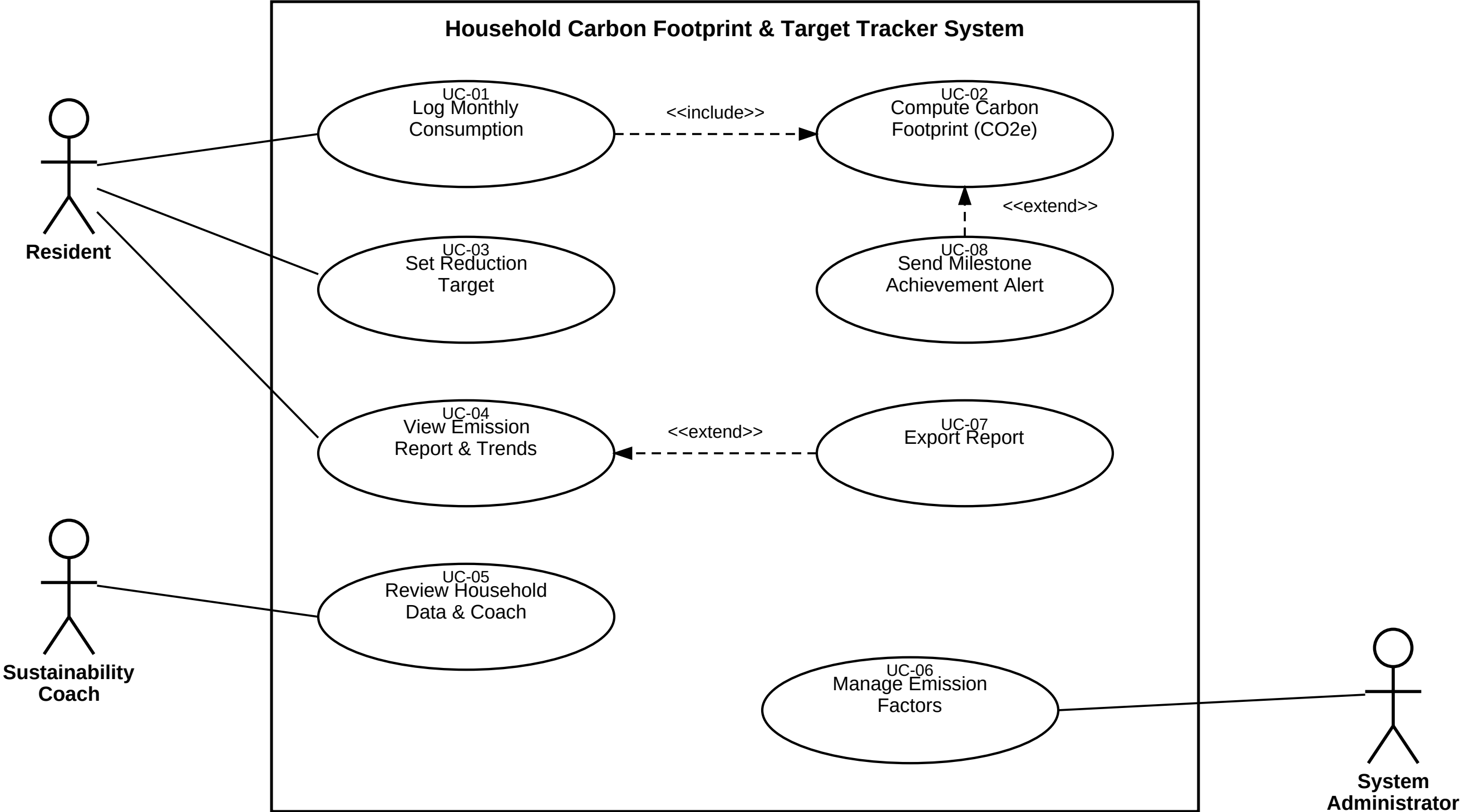


Requirements Table — Household Carbon Footprint & Target Tracker

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO ₂ e) emissions.	High	Pass: Calculated emissions match EPA conversion factor formulas. Fail: Zero or negative emission values generated.	Establishes the core data-ingestion and calculation capability that every other footprint-tracking feature depends on. (Given)
FR-002	Functional	The system shall allow a Resident to log monthly natural-gas consumption (therms or m ³) alongside electricity and transport data for a given household.	High	Pass: A submitted gas reading is stored against the correct household/month and included in the CO ₂ e total. Fail: Gas entries are excluded from the monthly calculation.	Gas heating and cooking are a significant emissions source; omitting them would make the footprint inaccurate.
FR-003	Functional	The system shall allow a Resident to define and edit a personal monthly or annual carbon-reduction target, expressed in kg CO ₂ e.	Medium	Pass: A saved target is retrievable and used to compute a progress percentage. Fail: No target can be saved, or a negative/zero target is accepted.	Implements the “Target Tracker” capability that is central to the problem statement, giving residents a concrete goal.
FR-004	Functional	The system shall display a historical trend chart comparing a household’s actual monthly CO ₂ e emissions against its reduction target for the past 12 months.	High	Pass: Chart renders a correct data point for each of the last 12 months and visually flags months where the target was missed. Fail: Chart omits months or plots incorrect values.	Visual feedback on progress is the primary mechanism for behaviour change and milestone recognition.
FR-005	Functional	The system shall allow a Sustainability Coach to view the emission history of any household assigned to them and submit written coaching recommendations.	Medium	Pass: A coach’s note is saved, timestamped, and visible on the Resident’s dashboard. Fail: The note fails to save or is not visible to the Resident.	Supports the Sustainability Coach actor and the collaborative, advisory purpose of the tool.
NFR-001	Nonfunctional (Performance)	The system shall render the historical emission comparison chart within 200 ms after the requested data is available.	High	Pass: In 20 test runs, at least 95% of chart renders complete within 200 ms. Fail: Any test run exceeds the threshold for more than 5% of requests.	Fast, responsive charts keep residents engaged; slow dashboards discourage regular logging. (Given, refined)
NFR-002	Nonfunctional (Security & Privacy)	All household utility-consumption data shall be encrypted at rest (AES-256) and in transit (TLS 1.2+), and accessible only to the household’s own Resident account and its explicitly assigned Sustainability Coach.	High	Pass: A security audit confirms encryption at rest/in transit and unauthorized accounts receive an access-denied response. Fail: Any unauthorized account can read another household’s data, or data is stored unencrypted.	Utility data reveals sensitive lifestyle patterns (occupancy, travel habits); protecting it is essential for user trust and privacy compliance.

Use-Case Diagram — Problem #61: Household Carbon Footprint & Target Tracker



Use-Case Flow Specification

Use Case:	UC-01: Log Monthly Consumption (includes UC-02: Compute Carbon Footprint)
Primary Actor:	Resident
Secondary Actors:	None

Preconditions

- The Resident is authenticated, linked to a household profile, and authorized to create or update the selected month's consumption entry.

Postconditions

- Success: A new monthly consumption entry (electricity, gas, transport) is stored, the household's CO₂e total for that month is computed and saved, the dashboard/trend chart reflects the new value, and a milestone alert is triggered if the reduction target was met.
- Failure: No entry is persisted and the Resident is shown a specific error describing what to correct.

Main Success Scenario

- 1. Resident selects "Log Consumption" from the household dashboard.
- 2. System displays a data-entry form for the target month, prompting for electricity (kWh), gas (therms/m³), and commute distance (km).
- 3. Resident enters the readings for the month and submits the form.
- 4. System validates that all values are numeric, non-negative, and within a plausible range for the household.
- 5. System invokes the «include» use case Compute Carbon Footprint, applying EPA emission-factor formulas to the electricity, gas, and transport figures.
- 6. System stores the computed monthly CO₂e total against the household and month.
- 7. System updates the dashboard and historical trend chart with the new monthly total.
- 8. System compares the updated monthly CO₂e total with the applicable reduction target for the selected target period.
- 9. If the target is met for the month, the system triggers the «extend» use case Send Milestone Achievement Alert, notifying the Resident.
- 10. Use case ends successfully.

Alternate Flow — 4a. Invalid Input

- 4a1. System detects a non-numeric, negative, or implausible value in one of the fields.
- 4a2. System displays a validation error next to the offending field and does not submit the entry.
- 4a3. Resident corrects the value and resubmits the form.
- 4a4. Flow resumes at Main Success Scenario step 4.

Alternate Flow — 3a. Duplicate Entry for the Month

- 3a1. System detects that a consumption entry already exists for the selected month.
- 3a2. System asks the Resident to confirm whether to overwrite the existing entry.
- 3a3. If confirmed, flow resumes at step 4 and the prior entry is replaced; if declined, the use case ends without changes.